Development of an AI-Based Diagnostic Method for Malaria Diagnosis from Thick Blood Smears

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Last Meeting

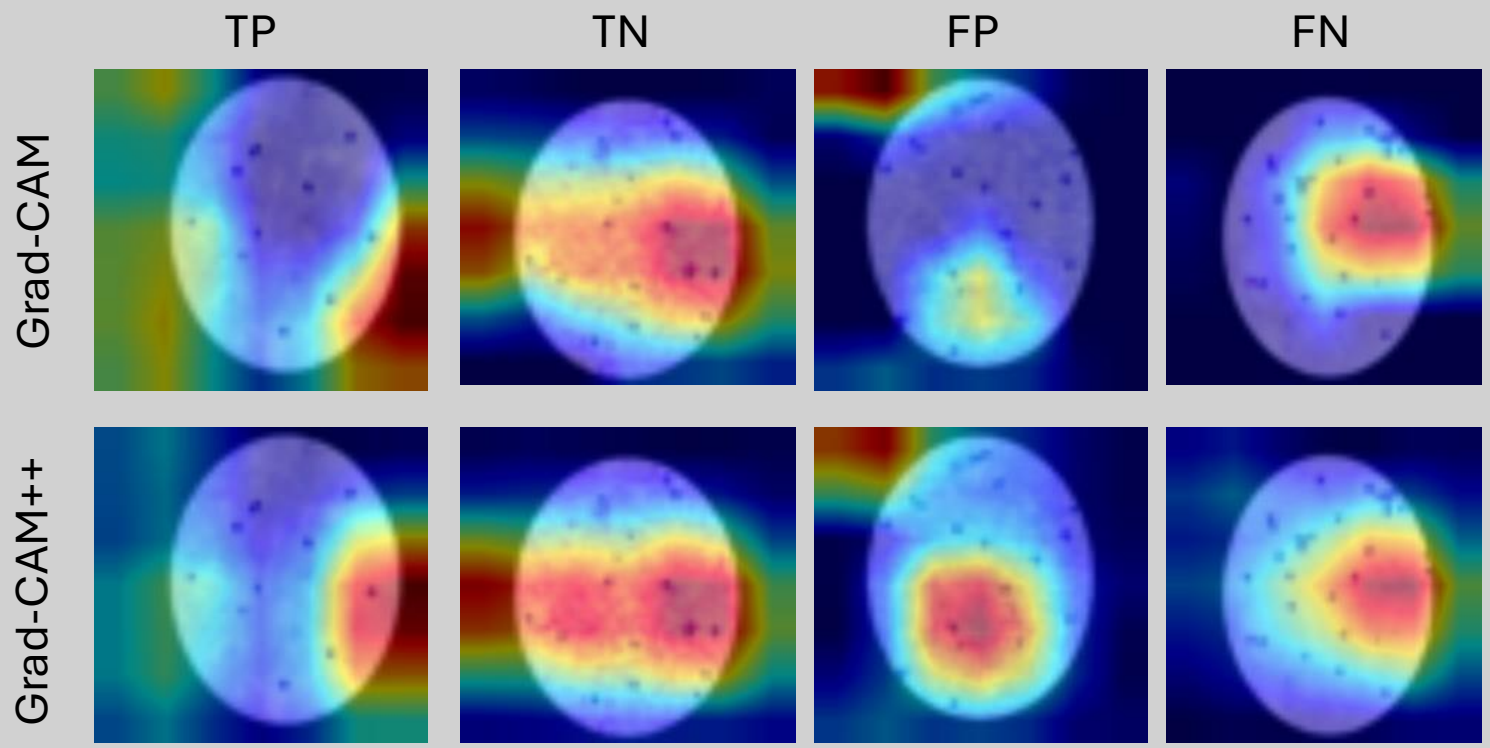
- ResNet50, LR3: attention on backgroundand corners.
- Specific to ResNet50, or general?

Today

- Grad-CAM, ResNet18
- MobileNet optimisation
- Grad-CAM, MobileNet
- Edges or background?
- External test (Uganda)

Grad-CAM on ResNet18 (LR3)

Grad-CAM and Grad-CAM++



- **Same pattern as ResNet50**
- **Attention on border and dark background, mostly TP and TN**
- **FP and FN: more attention inside the field**
- **Grad-CAM++: more focused**

Why optimise MobileNet?

- **Lightest model best fit for a smartphone**
- **But weakest at LR3(last meeting)**
- **Goal: improve it without changing the architecture**

MobileNet baseline, LR3

Accuracy	**89.30%**
Sensitivity	**94.80%**
Specificity	**68.97%**

MobileNet optimisation (LR3)

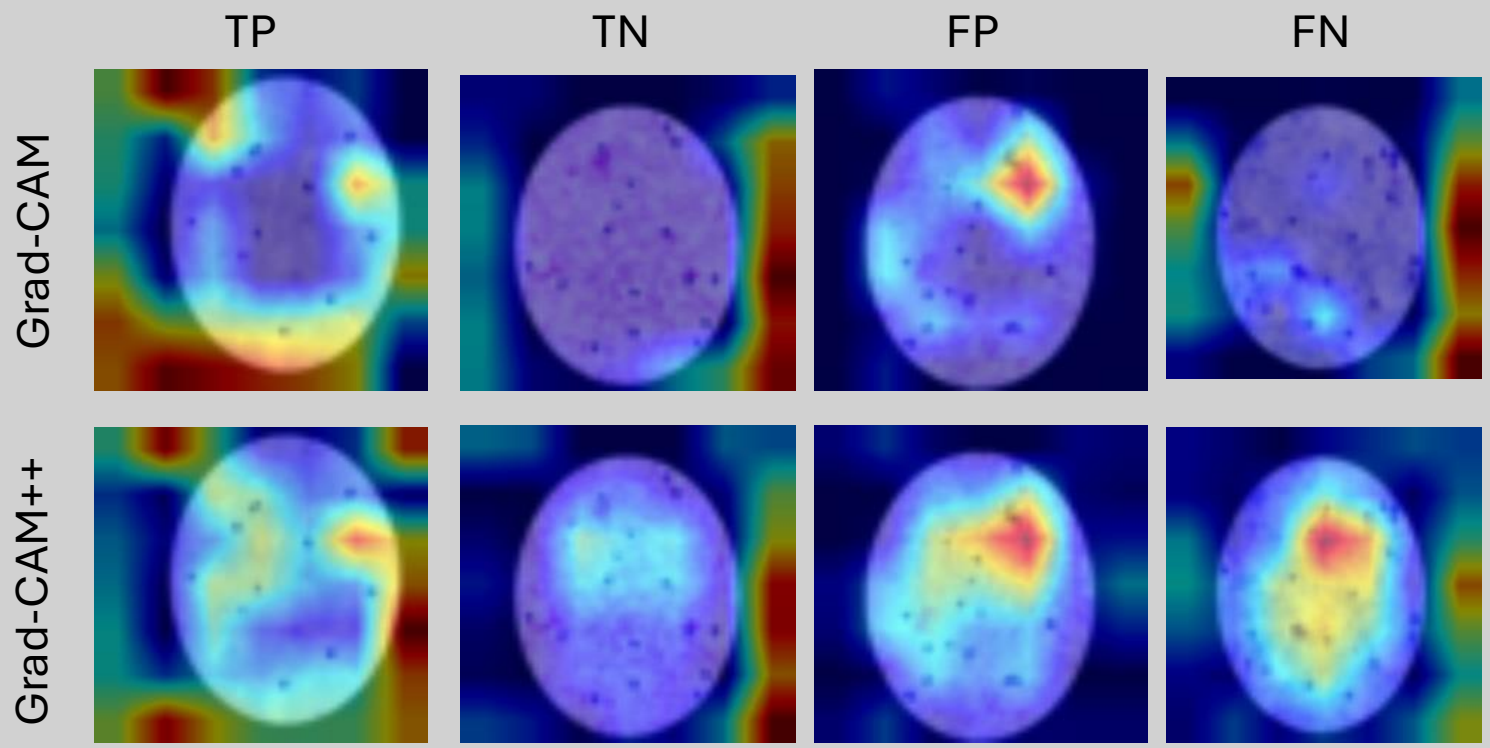
Only the training settings changed, the architecture stays the same

Stage	Training settings	Accuracy	Sensitivity	Specificity
Baseline	LR 0.0001, batch size 8,15 epochs	**89.30%**	**94.80%**	**68.97%**
Stage 1	LR 0.00003, batch size 16,30 epochs	92.34%	**99.20%**	67.00%
Stage 2	Stage 1 + class weights(uninfected weighted more)	**92.76%**	**96.40%**	**79.31%**

Test set: 953 images. Stage 2: most balanced result, specificity +10 points vs. baseline.

Grad-CAM on MobileNet (LR3)

Optimised model (stage 2), Grad-CAM and Grad-CAM++



What I see:

- **Activation around the circularborder, as in the other models**
- **FN: mostly on the edges**
- **TN: outside the field or atthe boundary**
- **FP: more inside the field**
- **TP: inside the field andat the edges**
- **Grad-CAM++: same pattern**

The question and the decision

What I see
All three models look at the borderand the dark background

The question:

Do the models rely on edges and background and not on the parasites?

My test set cannot answer this

Same dataset, same borderand background

Image: NLM
(Training)

Image: AIR Uganda
(new)

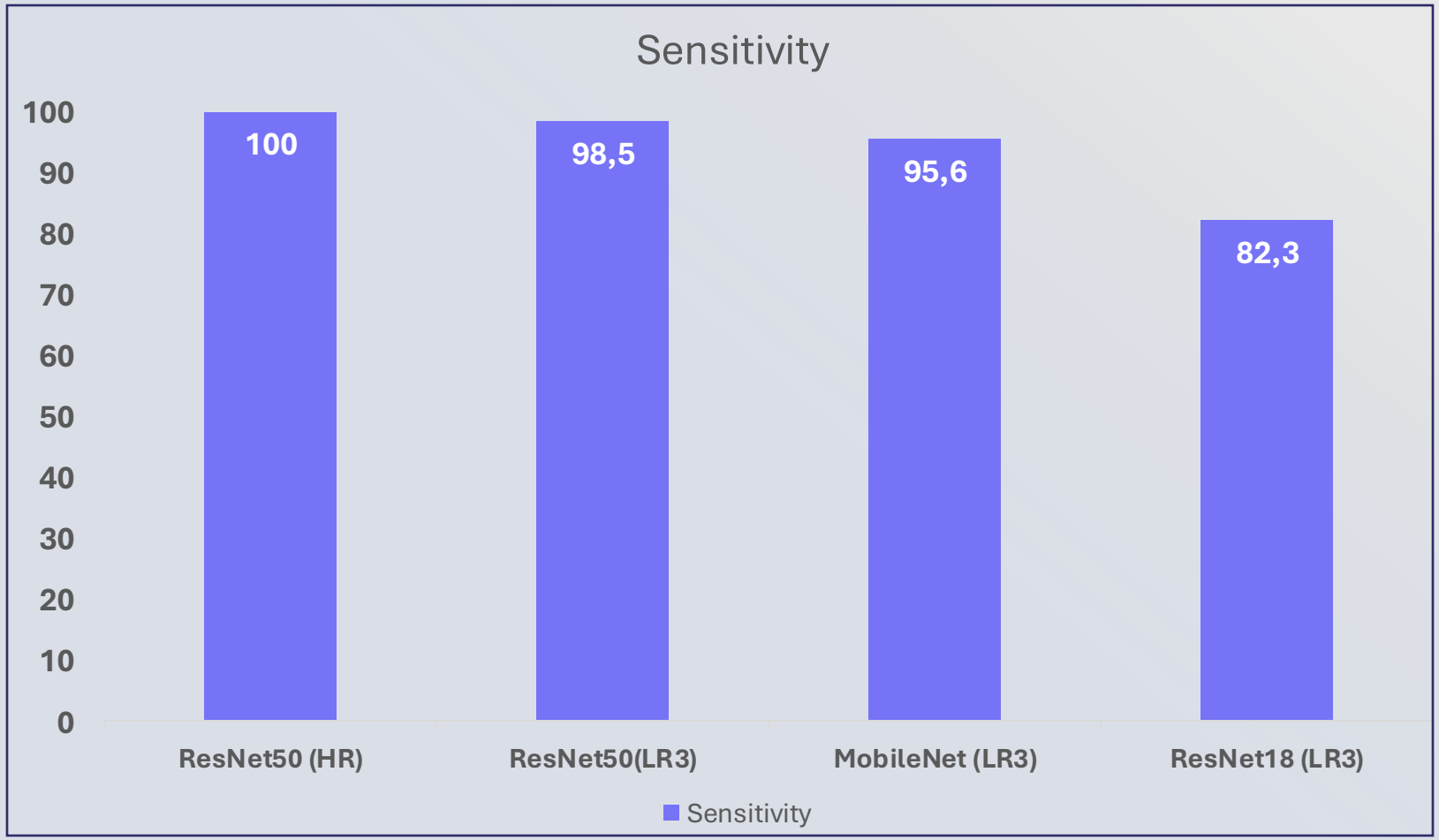
Decision: external test

- Different camera, no circular border
- 1182 images, 7245 parasites
- Infected only, so sensitivity only

4 models tested:

- ResNet50 (HR and LR3)
- MobileNet (LR3, optimised)
- ResNet18 (LR3)

External results (Uganda)



What it shows:

ResNet50 and MobileNetstill find almost all parasites

Even at LR3, on images from anew camera, with no border

ResNet18 is clearly weaker(82.3%)

What it means, and its limits

What it means:

- The models still find parasiteson new images
- So they do not depend only onthe border and the background
- ResNet50 and MobileNet alsowork at low resolution (LR3)

Limits:

- Infected images only,so no specificity
- Not proof that the border playsno role at all
- ResNet18 is weaker,reason not known yet
- MobileNet stage 2 was picked on the test set

Next step

Finish the thesis documentation

Thank you for your attention

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